Review plan · for the teacher

Best choices: unit check

Optimization · Level 1 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- What makes a choice the best?
- The shortest way
- Quick rules and their traps
- The best for whom?

Activity: Decision detectives

1. Warm-up (5 min): Collect real school decisions on the board: the seating plan, the duty rota, the sports day order, the lunch queue.
2. Three parts (10 min): Each group takes one decision and writes its options, its limits and its goal in three columns.
3. Quick rule (10 min): Groups make a plan with one quick rule (such as first come, first served or nearest first) and check it keeps every limit.
4. Fair? (10 min): Groups find who comes off worst in their plan and suggest one change that would be fairer, even if the goal score drops a little.
5. Share (5 min): Each group shares its decision, its plan and one improvement. The class votes on the fairest idea.

Use made-up names if a real decision is sensitive. The aim is to practise naming options, limits and goals, and asking who pays.

Discussion

- Which idea from this unit will you use first?
- Which quick rule do we trust too much at our school?
- When is a slightly lower score worth it for fairness?

Exit ticket

- One decision I will plan differently, and how.

Try it in class tomorrow

Pick one decision your school makes this month (a rota, the trip, exam rooms). With a colleague, write its options, limits and goal, try one quick rule, and ask who the plan is hard on.

Quiz sheet · for each team

Best choices: unit check

Optimization · Level 1 · Unit check

Team _____ Date _____

1 **Amina's class has 1,200 coins and five outing ideas, and wants the most votes. What is the limit?**

◯ The most votes ◯ The budget of 1,200 coins ◯ The five outing ideas

2 **A feasible plan keeps every limit, so it is always the optimal plan.**

◯ True ◯ False

3 **Which of these are goals: one number to make as large or as small as possible? Choose all that apply.**

☐ The least fuel for the bus round ☐ At most 1,000 kg on the truck ☐ No teacher in two classes at once
☐ The shortest walk to the clinic ☐ The most votes for the outings

4 **Put in order what happens on the village map with Dijkstra's method.**

☐ Bora, Rubi and Kesi are settled, and Mina drops to 9
☐ The school is settled: the shortest way is 13 km
☐ Tala is settled, and Kesi gets 7
☐ Home gets 0; Tala gets 4, Bora 5 and Rubi 6
☐ Mina is settled, and the school gets 13

5 **The bridge between Kesi and Mina floods. What is the shortest way to school now?**

◯ Through Rubi and Sana: 14 km ◯ Through Tala, Kesi and Mina: 13 km ◯ Through Bora: 15 km

6 **Match each quick rule to the trap it fell into in this unit.**

Nearest village next	• • • •	43 votes instead of 45
Most points per kilo first	• • • •	A total of 29 instead of 39
Favourite first	• • • •	34.6 km instead of 28.6 km
First come, first served	• • • •	60 points instead of 65

7 **Complete the sentence.**

Three _____ took the bus round from 34.6 km to _____ km, and every village was still _____.

Word bank: visited · swaps · villages · 40.2 · 28.6

8 **Adding a fairness rule, such as "nobody gets the job nobody wants two weeks running", can lower the best score on the goal.**

◯ True ◯ False

9 **A plan shares classes among teachers. Which numbers should you look at to judge whether it is fair? Choose all that apply.**

☐ The lowest score anyone gets ☐ The gap between the best and the worst
☐ How many teachers chose before lunch ☐ The total of all the scores

10 **The best plan for teachers and classes totals 39, against 29 for first come, first served. Who gives something up in the best plan?**

◯ Karim: he loses maths ◯ Amina: from maths (9) to history (6) ◯ Nobody: everyone gains

Answers · for the teacher

Best choices: unit check

Optimization · Level 1 · Unit check

1 **The budget of 1,200 coins**

The budget must hold in every plan, so it is the limit.

2 **False**

Feasible only means allowed. There are usually many feasible plans; the optimal one is the best of them on the goal, like 45 votes instead of 43.

3 **The least fuel for the bus round; The most votes for the outings; The shortest walk to the clinic**

Fuel, votes and walking distance are numbers to make small or large: goals.

4 **Top to bottom: 3, 5, 2, 1, 4**

Each turn settles the waiting place with the smallest number, then its roads lower its neighbours' numbers, until the school is settled at 13 km.

5 **Through Rubi and Sana: 14 km**

Without the bridge the 13 km way is gone.

6 **Nearest village next — 34.6 km instead of 28.6 km**
Most points per kilo first — 60 points instead of 65
Favourite first — 43 votes instead of 45
First come, first served — A total of 29 instead of 39

Every quick rule in this unit gave a feasible plan fast, and every one missed the best.

7 **Three swaps took the bus round from 34.6 km to 28.6 km, and every village was still visited.**

Each 2-opt swap reverses a stretch of the round to remove a crossing.

8 **True**

A new rule only removes plans, so the best score can stay the same or fall.

9 **The total of all the scores; The lowest score anyone gets; The gap between the best and the worst**

The total shows how good the plan is overall; the lowest score and the gap show who carries the cost.

10 **Amina: from maths (9) to history (6)**

Karim gains most (from science, 2, to maths, 9), but Amina gives up her favourite and drops from 9 to 6.